

20/2/2021

Lecture 14 (BCH Codes, Cyclic Codes)

Recap

Generalized Reed Solomon Codes.

→ definition using the parity check matrix

→ These codes are MDS.

→ Dual of GRS codes are GRS codes

→ GRS codewords can be viewed as polynomial evaluations.

→ Special case of GRS codes → RS codes.
by setting v_i' (column multipliers of generator matrix of GRS code) = 1.

History: Hamming Code (Richard Hamming 1950)

Reed Solomon Code (Irving Reed & Gustave Solomon → 1960)

BCH codes → Bose-Chaudhuri-Hocquenghem (BCH codes)

Reed Muller codes (Irving Reed & Gustave Solomon)

(Conventional) Reed Solomon Codes

→ Special Case of GRS Codes.

→ $n \mid (q-1)$. → There exists an element $\alpha \in \mathbb{F}_q$ such that
order of $\alpha = n$.
 $\alpha^n = 1$.

Because $\beta^{q-1} = 1$
 $n \mid q-1$

$\left(\beta^{\frac{q-1}{n}}\right)^n = 1$
 α that we are looking for
is $\alpha = \beta^{\frac{q-1}{n}}$.

→ $\alpha_j = \alpha^{j-1}$, $1 \leq j \leq n$ (Code locators).

→ $v_j = \alpha^{b(j-1)}$, $1 \leq j \leq n$ (Column multipliers).

$$H_{RS} = \begin{bmatrix} 1 & \alpha^b & \alpha^{2b} & \dots & \alpha^{(n-1)b} \\ 1 & \alpha^{b+1} & \alpha^{2(b+1)} & \dots & \alpha^{(n-1)(b+1)} \\ \vdots & & & & \\ 1 & \alpha^{b+n-k-1} & \alpha^{2(b+n-k-1)} & \dots & \alpha^{(n-1)(b+n-k-1)} \end{bmatrix}$$

Generator polynomial of $\mathbb{C}_{RS}$

$\underline{c} = (c_0, \dots, c_{n-1}) \in \mathbb{F}_q^n$ is a
codeword of $\mathbb{C}_{RS}$.

$$H_{RS} \cdot \underline{c} = 0.$$

$$\sum_{j=0}^{n-1} c_j (\alpha^l)^j = 0. \quad l \in \{b, b+1, \dots, b+n-k-1\}.$$

(n-k) equations.

Consider the polynomial $c(x) = \sum_{j=0}^{n-1} c_j x^j$
 α^l is a root of $c(x)$ $\uparrow$ code polynomial
 $\forall l = b, b+1, \dots, b+n-k-1$
 $(x - \alpha^l) \mid c(x).$ $\forall l = b, b+1, \dots, b+n-k-1$

$$\prod_{l=b}^{b+n-k-1} (x - \alpha^l) \mid c(x).$$

generator polynomial of
the RS code.

$$g(x) = \prod_{l=b}^{b+n-k-1} (x - \alpha^l).$$

$$\underline{\underline{g(x) \mid c(x)}}.$$

$$c(x) = i(x) g(x) \quad \text{for } i(x) \in \mathbb{F}_q[x].$$

$$\deg(i(x))$$

$$\deg(g(x)) = n - k.$$

$$\deg(c(x)) \leq n-1 \quad \leftarrow \quad \underbrace{c_0 + c_1 x + \dots + c_{n-1} x^{n-1}}$$

$$\deg(i(x)) \leq n-1 - \deg(g(x))$$

$$\leq k-1.$$

$$g(x) = \prod_{l=b}^{b+n-k-1} (x - \alpha^l).$$

Take arbitrary $i(x)$ with degree at most $k-1$ & multiply with $g(x)$ to get code polynomial

Polynomial

Evaluations

$u(x) \rightarrow$ Message polynomial
 $(u(d_1) \dots u(d_n))$

Generators

Polynomial based.

$c(x) = g(x) i(x).$
 $(c_0 \ c_1 \dots c_{n-1}) \dots$

Cyclic Codes $\rightarrow$ (CRC $\rightarrow$ Cyclic Redundancy
check)

Defn A linear code over $\mathbb{F}_q$ is said to be cyclic if every cyclic shift of a codeword is also a codeword.

$$(c_0, \dots, c_{n-2}, c_{n-1}) \in \mathbb{C}$$

$$\Rightarrow (c_{n-1}, c_0, c_1, \dots, c_{n-2}) \in \mathbb{C}.$$

In the codeword polynomial

$$c(x) = c_0 + c_1 x + \dots + c_{n-1} x^{n-1}$$

$$x c(x) = c_0 x + c_1 x^2 + \dots + c_{n-1} x^n$$

$$\underline{x c(x)} \pmod{x^n - 1}.$$

$$x c(x) \pmod{x^n - 1}$$

$$= c_{n-1} + c_0 x + c_1 x^2 + \dots + c_{n-2} x^{n-1}$$

$$c(x) \in \mathbb{C} \Rightarrow a(x) c(x) \pmod{x^n - 1}$$

$\in \mathbb{C}$
follows from linearity of the code.

$$a(x) = a_0 + a_1 x + a_2 x^2 + a_3 x^3.$$

$$c(x) = c_0 + c_1 x + \dots + c_{n-1} x^{n-1}$$

$$a_0 c(x) \rightarrow a_0 c_0 + a_0 c_1 x + \dots + a_0 c_{n-1} x^{n-1} \in \mathbb{C}.$$

$$a_1 x c(x) \rightarrow a_1 c_{n-1} + a_1 c_0 x + \dots + a_1 c_{n-2} x^{n-1} \in \mathbb{C}.$$

(mod $x^n - 1$)

$$a_2 x^2 c(x) \rightarrow a_2 c_{n-2} + a_2 c_{n-1} x + a_2 c_0 x^2 + \dots + a_2 c_{n-3} x^{n-1} \in \mathbb{C}.$$

(mod $x^n - 1$)

$$a(x) c(x) \pmod{x^n - 1} \in \mathbb{C} \rightarrow \text{Cyclic code.}$$

(right shifts +
scalar multiplication
+ addition)

Construction of Cyclic Codes

Let $g(x)$ be a (monic) polynomial that divides $(x^n - 1)$ in $\mathbb{F}_q[x]$. Let $\mathbb{C}$ consist of all polynomials of deg at most $n-1$ s.t. $g(x) \mid c(x)$. Then $\mathbb{C}$ is a cyclic code of blocklength n .

Proof that above construction results in a cyclic code.

① Linear.

if $g(x) \mid c(x)$ and $g(x) \mid d(x)$.

$$g(x) \mid (\alpha c(x) + \beta d(x))$$

$$\alpha c(x) + \beta d(x) \in \mathbb{C}.$$

② Closure under right shift operation.

$$\text{Let } b(x) = x c(x) \bmod (x^n - 1).$$

$$b(x) = x c(x) + a(x)(x^n - 1).$$

$$= x u(x) g(x) + a(x)(x^n - 1) \quad \left(\begin{array}{l} \text{Because} \\ g(x) \mid c(x) \end{array} \right)$$

$$= x u(x) g(x) + a(x) h(x) g(x). \quad g(x) \mid (x^n - 1)$$

$$= (x u(x) + a(x) h(x)) g(x).$$

$$\Rightarrow g(x) \mid b(x). \text{ Also note that } \deg(b(x)) \leq n-1$$

Corollary 1 :- Conventional Reed Solomon Codes are cyclic.

$$g(x) = \prod_{l=b}^{b+n-k-1} (x - \alpha^l)$$

To argue that $g(x) \mid (x^n - 1)$

We picked α of order n , $\alpha^n = 1$.

In particular, all α^l are roots of $x^n - 1$

$$(x - \alpha^l) \mid (x^n - 1) \quad \forall l = b, b+1, \dots, b+n-k-1$$

& hence.

$$\prod_{l=b}^{b+n-k-1} (x - \alpha^l) \mid (x^n - 1)$$

$$\Rightarrow g(x) \mid (x^n - 1).$$

Corollary 2 :- BCH Codes are also cyclic.

Because BCH Codes are subfield subcodes of RS Codes which themselves are cyclic codes.

Generator matrix of Cyclic Codes.

(n, k) cyclic code

$$\deg(g(x)) = n - k$$

$$c(x) = u(x)g(x)$$

from the generator polynomial

$$g(x) = g_0 + g_1x + g_2x^2 + \dots + g_{n-k}x^{n-k}.$$

$g(x)$ itself is a codeword & so are all its cyclic shifts

$$g(x), \quad xg(x) \pmod{x^n - 1}$$

$\vdots$

$$x^{k-1}g(x).$$

$$G = \begin{bmatrix} g_0 & g_1 & \dots & g_{n-k} \\ 0 & g_0 & \dots & g_{n-k} \\ & & \ddots & \\ & & & g_0 & \dots & g_{n-k} \end{bmatrix}$$

→ Dual of a cyclic code is a cyclic code.